Lesson 19: Trigonometric Identities — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Simplify $1 - \cos^2\theta$.

2. Simplify $1 - \sin^2\theta$.

3. Write $\tan\theta$ in terms of sin and cos.

4. Simplify $\sin^2\theta + \cos^2\theta$.

5. State $\sin 2\theta$.

6. State one form of $\cos 2\theta$.

Section 2 — Core practice

7. Given $\sin\theta = 4/5$ (acute), find $\cos\theta$.

8. Given $\cos\theta = 5/13$ (acute), find $\sin\theta$.

9. Given $\sin\theta = 3/5$, $\cos\theta = 4/5$, find $\sin 2\theta$.

10. Simplify $(\sin\theta/\cos\theta)\cdot\cos\theta$.

11. Given $\cos\theta = 8/17$ (acute), find $\sin\theta$.

12. Simplify $2 \sin \theta \cos \theta$.

Section 3 — Challenge & exam-style

13. Given $\sin \theta = 12/13$ and θ acute, find $\cos 2\theta$.

14. Given $\cos \theta = 3/5$ (acute), find $\sin 2\theta$.

15. Show that $(1 - \cos^2\theta)/\sin\theta = \sin\theta$.

16. Given $\tan \theta = 3/4$ and θ acute, find $\sin \theta$ and $\cos \theta$.